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EX NO: 3	**Spectrum Analysis of Sub-1 GHz, Mid-band s mmWave by Plotting Propagation Loss.**

AIM:

To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Mid-band, and mmWave) in wireless communication systems and compare their performance over varying distances using MATLAB.

Procedure:

1. Define frequency bands: Sub-1 GHz (0.9 GHz), Mid-band (3.5 GHz), and mmWave (28 GHz).
2. Set communication distance range (e.g., 0.1 km to 5 km).
3. Use Free Space Path Loss (FSPL) formula to compute propagation loss:

$$PL(dB)=20\log_{10}(d)+20\log_{10}(f)+32.44$$

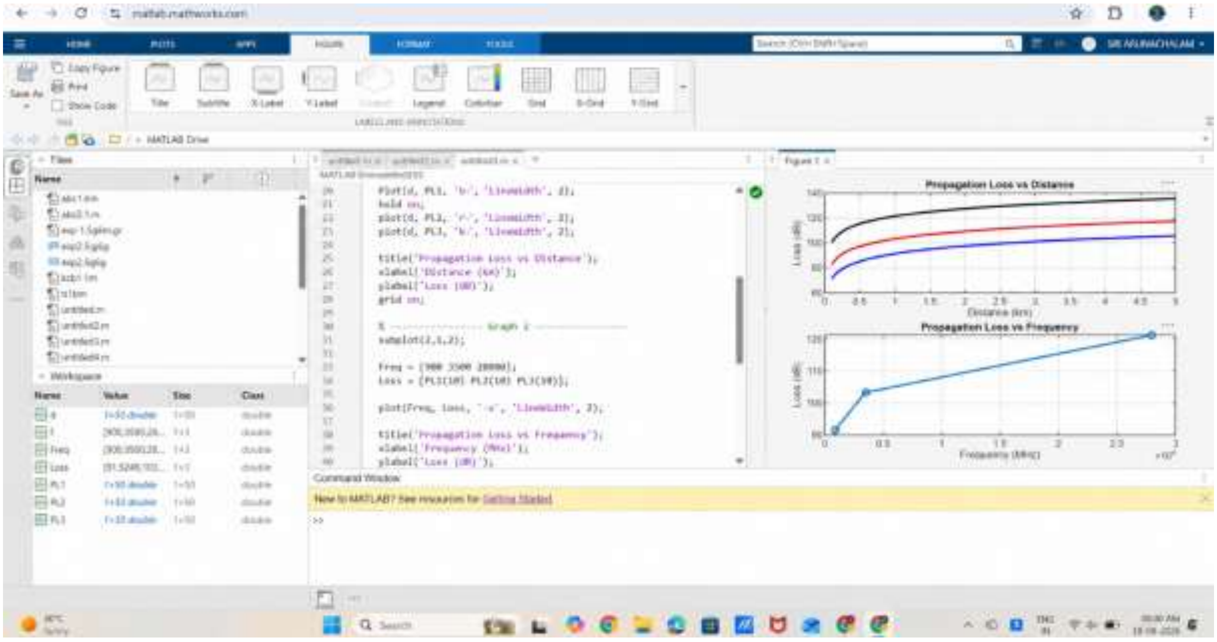
4. Calculate propagation loss for each frequency band over varying distances.
5. Plot distance vs propagation loss curves for comparison.
6. Analyze how higher frequencies experience higher attenuation.
7. Compare coverage efficiency of each spectrum band.
8. Display results using MATLAB plots.

Objective 1: Matlab Code and Output:

```
clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)

f = [900 3500 28000]; % Frequency (MHz)
```

```
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% ------ Graph 1 ------
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

OUTPUT:



Result:

The graph shows that propagation loss increases with both communication distance and operating frequency in wireless systems.